

Learning Beyond the IID Closed-Set Setting

Programming Assignment 1 — Draft (Tasks 1–3)

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Abstract

Draft compiled with Tasks 1–3 only. Task 4 is omitted from this build.

1 Task 1: Inductive Biases and Feature Representations

1.1 Experimental Setup

1.2 Clean Baseline vs. Visual Interventions

As summarized in Table 1, all three architectures maintain high clean accuracy, but their sensitivity diverges substantially under visual interventions...

1.3 Model’s Reliance on Shape, Texture, and Color

Model performance under interventions like color changes (greyscaling or hue modifications) showed no substantial drop in overall performance, suggesting that none of the models gave much importance to the color when making predictions. Removing color altogether cost between 1.4 and 2.4 points, and changing color impacted even less, from 0 to 1.6 points. Although, under the greyscale all four models broke more predictions than they managed to fix, only the zero-shot CLIP change was statistically more reliable. So we cannot claim that the color interventions meaningfully resulted in degraded performance under the given configuration and dataset.

Things did get a bit more interesting with the cue conflicts, where the texture of one class is pushed over the shape of another. Overall, all of the models showed a strong inclination towards shape bias, with ResNet being the least biased towards shape and appearing to give some importance to texture as well as compared to the rest. An important factor to highlight here is that the coverage of each model on the cue conflicts gives a different base for these percentages. As, for ResNet the shape bias is computed from 160 predictions, while for the other models it is computed from about 180. Coverage is the share of conflicts on which a model picked one of the two cued classes, so a lower coverage means the model committed to a cue on fewer images. The models, ViT, the CLIP head, and zero-shot CLIP, predicted a third class on only 17 to 20 of the 200 conflicts, while ResNet did so on 40. Because of this, ResNet’s shape bias is measured less precisely than the others, with a 95% range of 81.4 to 91.9 against ViT’s 95.6 to 99.5. So, we still cannot conclusively say whether ResNet’s lower coverage follows from its greater reliance on texture or from some other weakness, since a third-class prediction only shows that neither cue won, not which cue caused it.

With all of that, it is worth looking at whether texture contributed at all in models’ prediction flow and how strongly. Texture rank gives us that. Based on the texture-rank findings, for the three trained heads (ResNet +14.5, CLIP head +12.0, ViT +7.0 points)

Table 1: Top-1 accuracy (Acc, %), change relative to each method’s own clean baseline (Δ , percentage points) and prediction consistency (C, %) under the colour and patch-structure interventions. All four decision methods were evaluated on the same class-balanced subset of 500 STL-10 test images (seed 6304); consistency is the share of images whose predicted class is unchanged with respect to the clean image, so it counts predictions that stay the same even when both are wrong. Changes marked $\dagger$ are statistically reliable under a paired McNemar test with Holm correction within this family of twelve tests ($p < 0.05$); all remaining changes are within sampling noise. Hue rotation of 30° is the additional colour intervention, and patch shuffle uses a 4×4 grid of 56×56 pixel blocks with one fixed non-identity permutation per image.

Method	Clean	Grayscale			Hue 30°			Patch shuffle		
	Acc	Acc	Δ	C	Acc	Δ	C	Acc	Δ	C
ResNet-50 head	97.8	96.4	-1.4	97.0	96.2	-1.6	97.0	90.0	-7.8 $\dagger$	91.8
ViT-B/16 head	98.8	96.6	-2.2	96.6	98.0	-0.8	98.4	92.6	-6.2 $\dagger$	92.8
CLIP ViT-B/32 head	97.8	95.4	-2.4	95.6	97.8	+0.0	97.4	84.0	-13.8 $\dagger$	83.6
CLIP ViT-B/32 zero-shot	96.8	94.4	-2.4 $\dagger$	96.8	95.8	-1.0	98.4	84.0	-12.8 $\dagger$	84.4

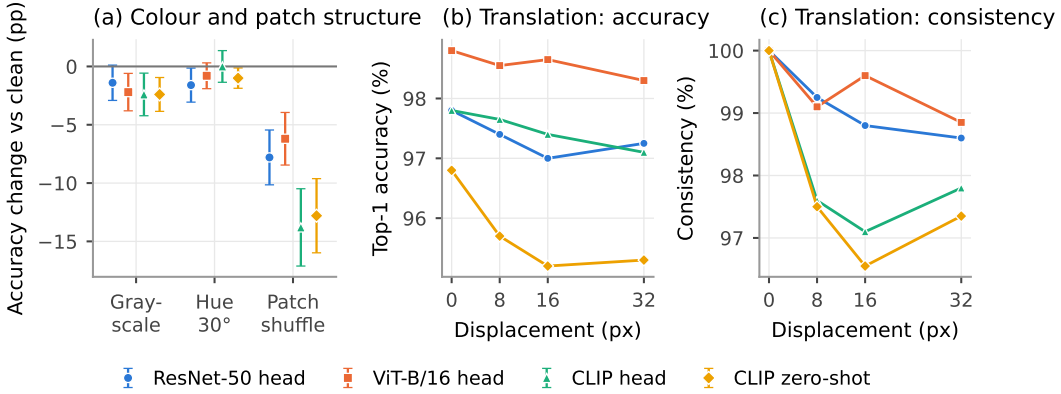


Figure 1: Effect of the controlled interventions on the four decision methods. (a) Change in top-1 accuracy relative to each method’s own clean baseline, in percentage points, with 95% confidence intervals for the paired change; the horizontal line marks no change, so an interval crossing it indicates a change that cannot be distinguished from sampling noise. The colour interventions cluster near zero while the patch-shuffle points lie well below it. (b) Mean top-1 accuracy and (c) mean prediction consistency against translation displacement, averaged over the four cardinal directions at each displacement; zero displacement reuses the clean evaluation and therefore has 100% consistency by construction. All panels use $N = 500$ images per condition.

the texture class rose into the top two of the ranked class scores more often under conflict than on the clean content image, so texture does influence their scores even though shape still dominates the final decision; zero-shot CLIP showed no such reliable movement.

Coverage reflects the fraction of cue-conflict images for which the model predicts either the correct shape or texture class, rather than defaulting to an uninformative third category. As shown in Table 2, ...

1.4 Spatial and Positional Invariance

Models show almost no drop in accuracy and performance with translations. Out of the 48 translation tests (8, 16, and 32 pixels across four directions and four methods), only 8 resulted in 10 or more predictions flipping to negative, with a maximum of 16 (zero-shot CLIP at 16 pixels left), and 7 of these 8 cases belong to zero-shot CLIP. Nevertheless, given just 500 evaluation images and only a few flipped predictions, this analysis lacks enough

Table 2: Cue-conflict decisions on the 200 selected conflict images. Each prediction is counted as the content (shape) class, the style (texture) class, or another class entirely. Shape bias is $N_{\text{shape}}/(N_{\text{shape}} + N_{\text{texture}})$ and coverage is $(N_{\text{shape}} + N_{\text{texture}})/N_{\text{total}}$, so the denominator of the shape-bias column is the count in brackets rather than 200. Square brackets give 95% intervals (bootstrap over conflicts for shape bias, Wilson for coverage). All four methods were evaluated on identical images. The conflicts were generated with AdaIN at a style strength of $\alpha = 0.7$ and retained only when the content shape remained recognizable, which raises the absolute shape-bias level for every method.

Method	Shape	Texture	Other	Decisions	Shape bias (%)	Coverage (%)
ResNet-50 head	139	21	40	160	86.9 [81.4, 91.9]	80.0 [73.9, 85.0]
ViT-B/16 head	179	4	17	183	97.8 [95.6, 99.5]	91.5 [86.8, 94.6]
CLIP head	171	11	18	182	94.0 [90.2, 97.2]	91.0 [86.2, 94.2]
CLIP zero-shot	172	8	20	180	95.6 [92.2, 98.3]	90.0 [85.1, 93.4]

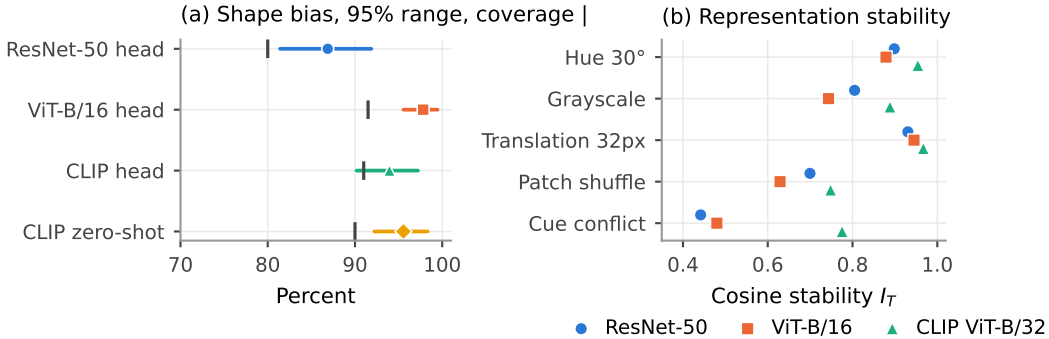


Figure 2: (a) Shape bias per decision method with its 95% interval (line), and coverage marked separately (|). The interval length reflects how many decisions the bias is computed from: ResNet-50 commits to a cue on 160 of 200 conflicts and therefore has a wider interval than ViT-B/16, which commits on 183. (b) Cosine stability I_T , the mean cosine similarity between the clean and transformed feature vectors of the same image, for each backbone and intervention; a value of 1 means the representation is unchanged. The translation row averages the four directions at 32 pixels. The CLIP head and zero-shot CLIP share one backbone and therefore one set of similarity values, so three backbones appear rather than four decision methods. $N = 500$ per condition, except cue conflict with $N = 200$.

statistical power to detect minor effects. The confidence intervals permit a performance decline of up to approximately 4 points, thus ruling out major degradation from translation while leaving smaller losses possible.

Things change with patch shuffling. All models suffered a substantial loss in accuracy, and this is consistent across models. All four drops were statistically reliable, with ResNet and ViT models’ performance declining by 6.2 to 7.8 points, where ResNet failed on 39 correct baseline predictions and was unable to correct a single incorrect baseline prediction. The CLIP models appeared to be hit the worst, with the CLIP head breaking 75 predictions while fixing only 6, and overall performance decreasing by 12.8 to 13.8 points. This difference is reliable, as both CLIP methods lost significantly more than ResNet and ViT.

Between translation and patch shuffling, the variation in model performance can be attributed to the different types of spatial changes introduced by each intervention. During the translation interventions, most of the global image and shape structure is preserved. Models still have access to enough global representation, evident from the feature similarity between 0.93 and 0.98 when comparing baseline and translated images, and they therefore retain their accuracy and consistency. This changes substantially with patch shuffling, which moves the patches around and destroys the global spatial arrangement: representational similarity falls to between 0.63 and 0.75, and all models lose accuracy. The correspondence holds

when comparing the two interventions, but not when comparing models, since CLIP keeps the highest similarity under shuffling (0.748) while losing the most accuracy (−13.8 points), and ViT has the lowest similarity (0.629) with the smallest loss (−6.2 points). Furthermore, local features retain significant diagnostic value independently: since patch shuffling keeps every pixel and local texture within each block intact, the models maintain an accuracy of 84.0% to 92.6% despite the loss of spatial layout—substantially above chance level. This demonstrates that local texture information sustains the majority of the classification performance, whereas global structure provides the final 6.2 to 13.8 percentage points.

1.5 Representation Space Analysis

The feature representation similarity for all models remains quite stable across translations when compared with the baseline. The similarity drops a little for the color interventions but still stays above 0.74. However, a substantial decline is observed for patch shuffling, to 0.63 for ViT, 0.70 for ResNet, and 0.75 for CLIP, and the cue conflicts caused the sharpest drop of all, down to 0.44 for ResNet and 0.48 for ViT, while CLIP still maintained a relatively high 0.78.

In the case of the translation interventions, the features barely move, staying between 0.93 and 0.98, and accuracy remains stable. The clearest divergence appears with the cue conflicts, where ResNet’s representation moves the furthest of all (dropping to 0.442), while its predictions change only slightly, since 139 of 200 still follow the shape class. A possible explanation is that cosine similarity measures how far the representation moved, not what it still contains, so a large movement does not necessarily remove the information the classifier relies on. In these cases, prediction stability and representation stability come apart.

Another interesting finding is observed for ViT on the cue conflicts, where the similarity is 0.505 for stable predictions against 0.264 for flipped ones, showing that the images whose predictions flipped had moved further than those that held. However, under patch shuffling that gap nearly vanishes (0.014 for ResNet), so feature movement barely explains which predictions broke there. Representation change and prediction change therefore agree within a condition, while diverging across conditions.

CLIP managed to keep the most stable representation under every intervention, yet it lost the most accuracy under shuffling (−13.8 points), while ViT moved furthest (0.629 under shuffling) and lost the least (−6.2 points).

CLIP vs CLIP Zero-Shot. While analysing the models’ dependency on feature representation, we can take the two CLIP methods, which share the same backbone and therefore the same learned image embeddings, one with a head trained on STL-10 and the other using zero-shot prompts, and compare their accuracies, their agreement, and their sensitivity to shape versus texture. Looking at accuracy, the two mostly agree, where agreement means both predicted the same class for the same image: they agree closely on the baseline, the translations and the colour changes, and their accuracy and agreement ratios are almost similar. The visible differences come from patch shuffle and cue conflicts. Under patch shuffle the accuracy numbers of the two are identical (84.0%), yet their agreement on the predicted class drops to 88.2%, meaning they disagreed on 59 of 500 images, with 22 correct only for the head and 22 correct only for zero-shot. Under cue conflicts agreement is similarly low at 88.0%, and the texture-rank analysis shows that the transferred texture reliably raised the texture class in the head’s top two (+12.0 points) but not in zero-shot’s (+0.5, not reliable). So even though both share the same backbone embeddings, the two decision rules read those same representations differently, and the trained head is the more texture-sensitive of the two.

1.6 Architecture and Observation Relationships

The results observed during the experimentation differ across the models. These models differ across their architecture, pretraining data, supervision, and capacity. As in the case of ResNet and ViT against CLIP. Such cross-model differences are therefore confounded and cannot be attributed to architecture alone.

Across this experimentation, the one causal claim that can be made about the decision rule comes from comparing CLIP with a trained linear head against zero-shot CLIP. Since both share the same frozen backbone, the same pretraining and identical image embeddings, the observed differences must be driven by the decision rule rather than the representation. This is visible under patch shuffle, where both reached the same accuracy of 84.0% yet disagreed on 59 of 500 images, and again on the cue conflicts, where the transferred texture reliably raised the texture class into the head’s top two (+12.0 points) but produced no reliable movement for zero-shot (+0.5).

A partial architectural link can be drawn between ResNet and ViT. Both were pretrained on the same dataset (ImageNet-1k) with the same supervision (class labels), so data and supervision are held constant. ResNet’s reliably lower shape bias (86.9% vs 97.8%) and lower coverage (80.0% vs 91.5%, driven by 40 third-class predictions concentrated in cases like monkey-to-truck and deer-to-car) indicate a relatively greater reliance on texture than the other three, which is consistent with the convolutional inductive bias reported in the literature. However, the two models also differ in capacity (25M vs 87M parameters) and training recipe (the IMAGENET1K_V2 ResNet weights use heavier augmentation), so architecture cannot be isolated as the cause. The shape bias built into our conflict set raises the absolute level for every method equally, so it explains the high values overall but not the difference between models.

Conversely, CLIP’s substantially larger accuracy degradation under patch shuffle (−12.8 to −13.8 points, compared with −6.2 to −7.8 for ViT and ResNet) cannot be assigned to architecture, since CLIP differs from ViT in patch size (32 px vs 16 px), supervision (multi-modal contrastive) and pretraining scale (400M web pairs) at the same time. The fact that both CLIP variants suffer equally does show that the effect comes from the representation rather than the decision rule.

Definitively isolating these underlying mechanisms would require controlled ablations holding non-architectural factors fixed, such as comparing a supervised ViT-B/16 directly against a CLIP-trained ViT-B/16, or training CNN and Transformer backbones under identical dataset sizes, augmentations, and optimisation recipes.

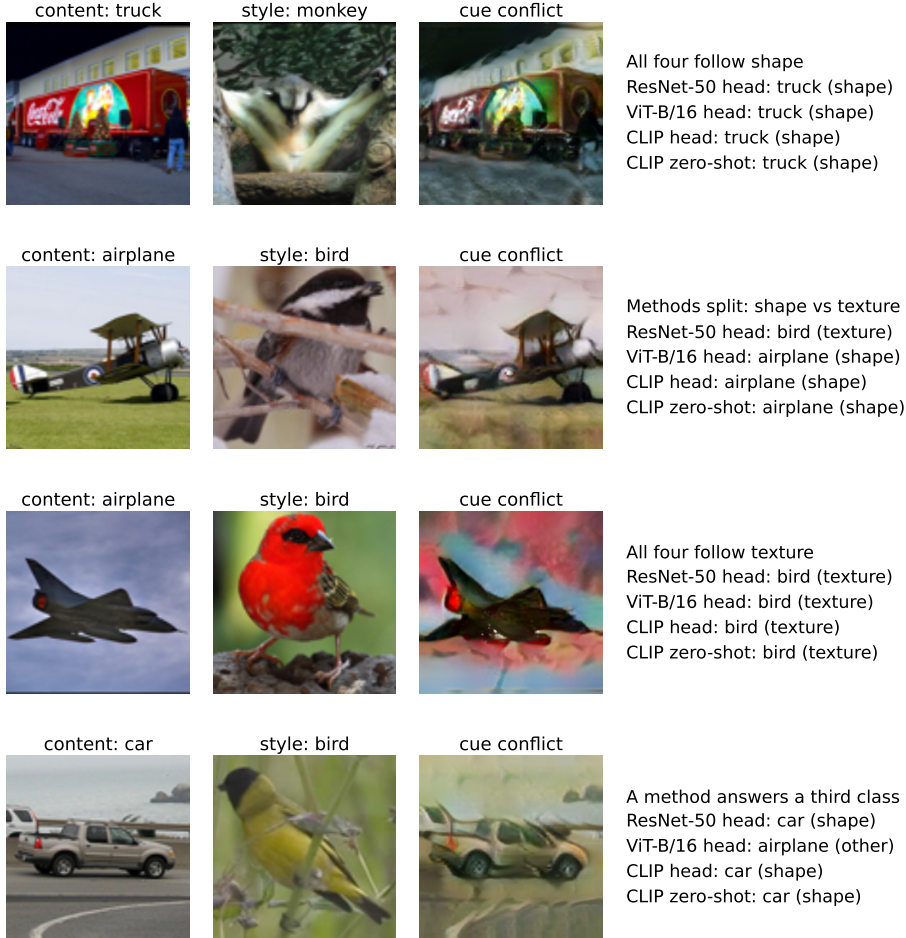


Figure 3: Representative cue-conflict images with the predictions of all four decision methods. Each row shows the content image supplying the shape, the style image supplying the texture, and the resulting conflict image, followed by each method's predicted class and whether that prediction counts as the shape, texture, or a third class. Rows illustrate the four decision patterns present in the evaluation set: unanimous shape decisions (126 of 200 conflicts), a split between shape and texture (24), unanimous texture decisions (2), and at least one method answering a third class (55). Examples were drawn with a fixed seed from each pattern, and model predictions played no part in selecting or retaining any conflict image.

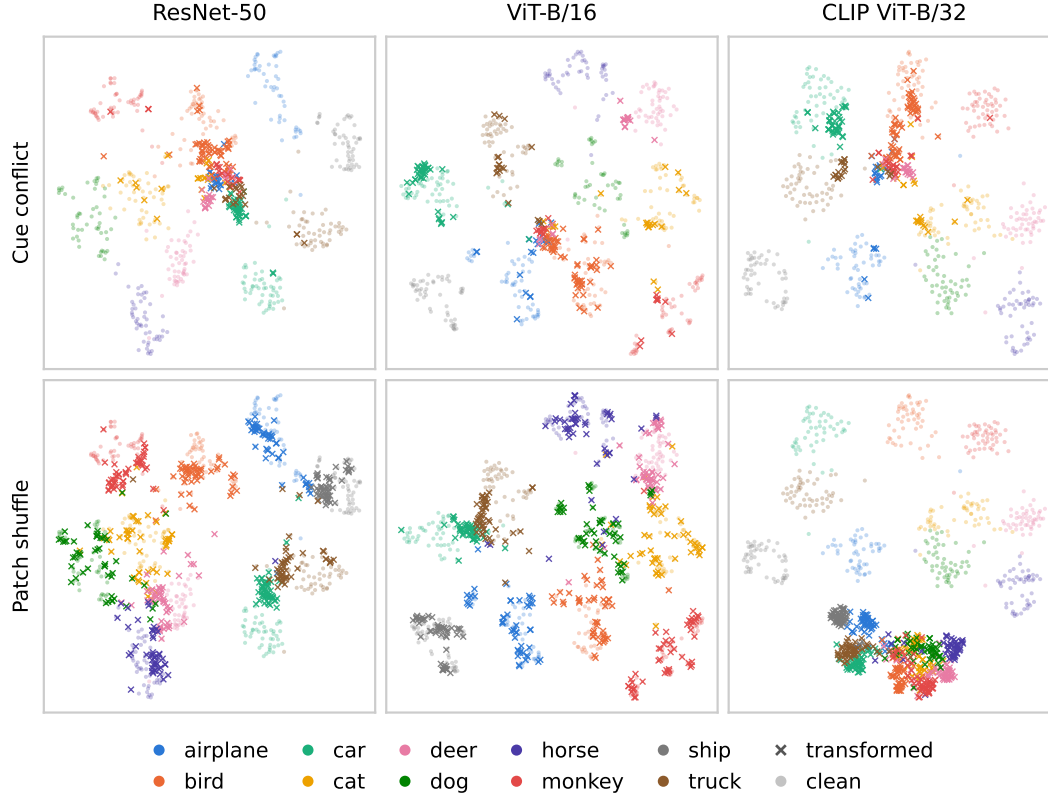


Figure 4: Joint t-SNE projections of clean and transformed representations, one fit per backbone over the combined clean and transformed features so that both conditions share a coordinate system. Columns are backbones and rows are interventions; colour identifies the ground-truth class (the content class for cue conflicts) and marker style separates clean from transformed examples. Features were L_2 -normalized and reduced with PCA before the projection (perplexity 30, seed 6304). Because each backbone is fitted separately, coordinates are comparable only within a panel column and never across backbones. Note the CLIP patch-shuffle panel, where the transformed points form a distinct region away from every clean cluster while the same model still classifies 84.0% of those images correctly.

Table 3: Source-only ERM and the three adaptation methods on PACS with Sketch as the unlabelled target. Source columns give macro-F1 on each held-out source validation split (a stratified 20% of each source domain, seed 6304) and their mean, which is the only quantity used for checkpoint selection. Target columns give accuracy and macro-F1 on the complete Sketch domain, evaluated once after every configuration and checkpoint was frozen, together with the change in target accuracy relative to Source-only ERM. Separability is the held-out accuracy of a balanced logistic-regression probe ($C = 1$, 70/30 split, seed 6304) trained to predict source versus target from frozen features, so 50% is chance and 100% means the two domains remain perfectly distinguishable. Accuracy and macro-F1 diverge where a method trades performance on large classes for performance on small ones; macro-F1 weights all seven classes equally.

Method	Source validation (macro-F1)				Target: Sketch			
	Photo	Art	Cartoon	Mean	Acc	Macro-F1	Δ Acc	Sep.
Source-only ERM	95.53	92.64	95.37	94.51	55.71	55.90	—	99.86
DAN (MMD)	94.88	88.85	94.56	92.76	67.80	66.84	+12.09	89.70
DANN ($\alpha_{\max} = 1$)	86.73	80.30	83.25	83.43	41.51	36.71	−14.20	100.00
CDAN	93.87	90.26	94.12	92.75	62.36	63.12	+6.64	98.49
<i>Controlled study: maximum gradient-reversal strength in DANN</i>								
DANN ($\alpha_{\max} = 0.25$)	96.58	91.31	94.61	94.17	28.23	20.40	−27.49	100.00
DANN ($\alpha_{\max} = 0.5$)	94.42	87.15	91.10	90.89	38.92	32.60	−16.80	99.86

2 Task 2: Unsupervised Domain Adaptation

2.1 Experimental Setup

2.2 The Source-Only Domain Gap and Its Failure Modes

The source-only ERM showed a substantial domain gap of 38.61 points (macro-F1) between the source and target domains. Although the model achieved a good 94.51% macro-F1 on the source domain, on the target (Sketch domain), it only attained 55.90%. The model showed a healthy decrease in training loss from 0.477 to 0.049, respectively observed an increase in training accuracy up to 98.79%, and a reflective increase in source-validation macro-F1, peaking at 94.51% at epoch 11. The performance remained consistently high across the source domains (Photo at 95.53%, Art Painting at 92.64%, and Cartoon at 95.37%). This demonstrates that ERM made the model learn class-discriminative source-domain features; however, these features are not directly transferable to the target domain and contain strong domain-specific information, which is clearly visible in the 99.86% domain-separability score.

The classes that resulted in the most important target failures were found by accounting for the overall error share of each class. As per these numbers, Giraffe, Dog, and Horse accounted for 86.7% of the overall errors on the target domain. Additionally, the model also displayed a strong bias towards the person class, and the model confused the majority of the classes with person, producing around 1,116 such errors. In contrast, the elephant and guitar classes showed good results, obtaining 88.78% and 83.06% accuracy. This shows that the domain shift did not affect each class uniformly.

2.3 Domain Separability and Target Recognition

After the linear probe via logistic regression, we got the scores reported in Table 3. Across all methods, we see a clear enough pattern: the higher the domain separability between source and target domains, the lower the target accuracy scores. ERM set the baseline for separability at around 99.86, with the target accuracy at 55.71 points. The treatment methods like DAN and CDAN were able to fractionally reduce this separability gap, and as the domain separability scores reduced with these methods, so did the accuracy on the

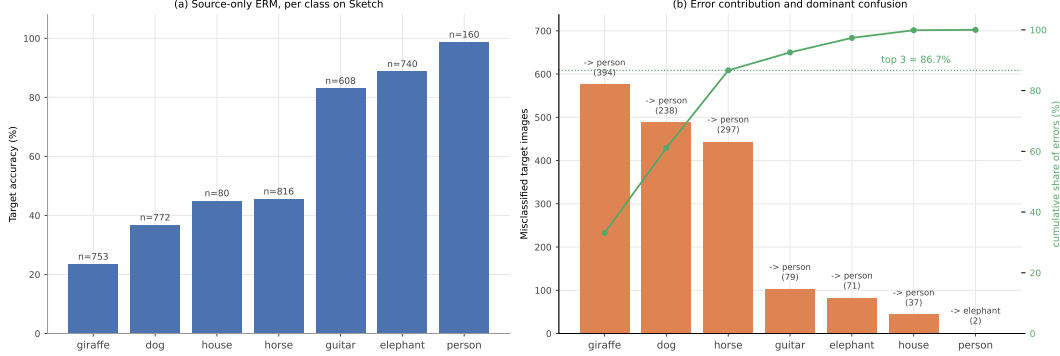


Figure 5: Where the source-only model’s target errors originate. (a) Per-class accuracy of Source-only ERM on the Sketch domain, ordered from weakest to strongest, with the number of target images of each class annotated above the bar. Support matters for interpretation: **house** and **horse** reach almost the same accuracy, but **house** has only 80 target images against **horse**’s 816. (b) The same result expressed as misclassified images, ordered largest first, with the cumulative share of all target errors overlaid (right axis). The dotted line marks the contribution of the three largest error sources. Each bar is annotated with the class that absorbs most of those errors and the number of such predictions, which shows that six of the seven classes send their dominant confusion to **person**.

target domain improve. DANN, though it performed miserably because of an unstable optimization flow, shows that it has the highest domain separability score and thus has the lowest target-domain accuracy.

The effect of domain separability is clearly visible in the class-level results as well. The methods that performed better compared to the others, like DAN or CDAN, showed successful adaptation. However, this improvement was not uniform across all of the classes. In DAN, the adaptation improved 5 out of 7 classes, with giraffe improving by 27.89 points, house by 55.00 points, and dog by 26.81 points. However, for elephants and persons, it exhibited negative transfer, declining in performance by 1.62 and 1.25 points. Similarly, CDAN also adapted successfully overall and improved 6 out of 7 classes, except elephant, where its performance decreased by 31.62 points. DANN displayed negative transfer overall. Despite showing adaptation on dog, giraffe, and house, it reported terrible performance on guitar, person, horse, and elephant. Its target macro-F1 plummeted by 19.19 points relative to ERM, most likely because of the instability it faced during optimization.

Overall, reduced domain separability can be associated with better performance on target recognition for DAN and CDAN. However, for good adaptation, per-class performance matters as well.

2.4 Class-Conditional versus Marginal Alignment

The results do not clearly support the claim that class conditioning helps preserve class semantic features better. The treatment methods DAN and DANN use two different approaches for domain alignment, where DAN uses MMD to compare distance between the source and target distributions and uses it as a loss penalty, i.e., the larger the distance between the source and target domain distributions, the greater the loss penalty, forcing the model to learn features that bring the two distributions closer in vector space so that the feature extractor learns about features that have resulted in a reduced gap between the distributions. With DAN, the domain separability score reduced down to 89.69 from 99.86 and showed the best target adaptation performance, reaching 66.84% macro-F1; however, the individual class-wise macro-F1 for DAN is not uniform across the classes, as it performed really well on 5 out of 7 classes (with giraffe +27.89, house +55.00, and dog +26.81 points), and for elephant and person, it saw a marginal decline. CDAN also observed an improved

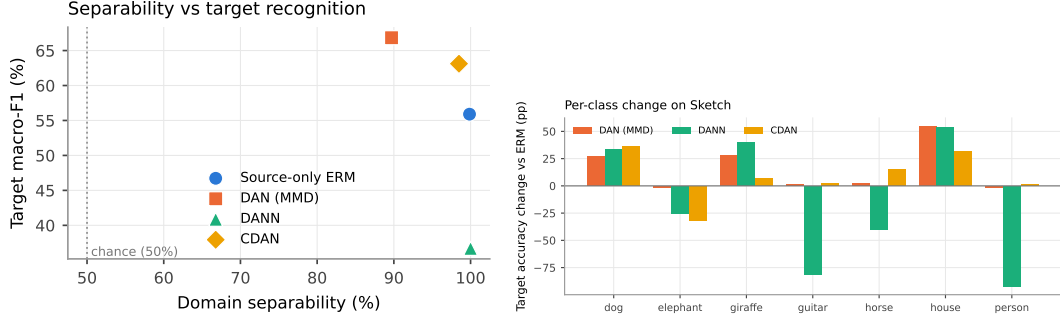


Figure 6: Aggregate and class-level evidence on whether reduced domain separability corresponds to improved target recognition. **Left:** target macro-F1 against domain separability for the four methods; the dotted line marks chance separability (50%). Only DAN moves appreciably away from the 100% ceiling, and it also achieves the highest target macro-F1, but CDAN improves on Source-only ERM while remaining at 98.49% separability, so reduced separability is not a precondition for successful adaptation. **Right:** change in per-class target accuracy relative to Source-only ERM, in percentage points, for each adaptation method. Bars above zero indicate positive transfer and bars below zero negative transfer. DAN and CDAN improve five and six of the seven classes respectively, while DANN’s aggregate collapse is concentrated in **guitar**, **person** and **horse**.

domain adaptation and reached a 63.12% target macro-F1, marginally less than DAN. Overall, it performed well on 6 out of 7 classes and also preserved person and guitar recognition, improving them by 1.25 and 2.47 points, but reduced the adaptation on the elephant class by a massive 31.62 points, and also provided just a 6.51-point gain on the giraffe, whereas on the same giraffe class DAN improved the adaptation by 27.89 points. Thus, the class conditionality might have partially preserved the class-wise consistency, with 6 classes improving; however, it did not manage to improve the overall aggregate for target adaptation.

CDAN does outperform the marginal adversarial alignment DANN, where DANN only reported 36.71% macro-F1. DANN also diverged with extremely larger classification and alignment losses and the optimizational instability, which is a key confounding factor in the underperforming DANN. Overall results may suggest that conditioning can better preserve semantic structure, and it improved aggregate target adaptation relative to ERM but did not outperform DAN, and stable DAN depicted a good marginal alignment that may outperform CDAN.

2.5 Effect of Increasing Alignment Strength

Overall baseline model performance with DANN, having alignment strength = 1.0, started well in the very first epoch. At the end of epoch 1, the classification loss was 0.61 (well below $\ln(7) = 1.946$), training accuracy reached 80.95%, and validation F1 reached 83.21%. Meanwhile, the alignment loss reached 2.78 (exceeding $\ln(2) = 0.693$), while domain accuracy settled at 58.18%. However, from epoch 2 onwards, the overall optimization became unstable, as the model’s classification loss shot up to 15.21 in the second epoch; the training accuracy dropped to 69.80%, while the validation F1 moved up to 83.43%, which tells us the model is making some very confident errors as its training accuracy decreases while its validation score remains high. Source-validation performance then collapsed outright from the 3rd epoch, when the validation macro-F1 plummeted from 83.43% down to 35.96% and then eventually reached 7.83%. Both the probabilistic loss and classification performance now show a total failure. The domain accuracy stayed between 42.29% and 58.18%; however, the alignment loss increased from 2.78 at the end of the first epoch to 97.49 in the second, reaching 2,013,773.19 in 7 epochs.

As we halve DANN’s alignment strength, things do start to get a little better for the

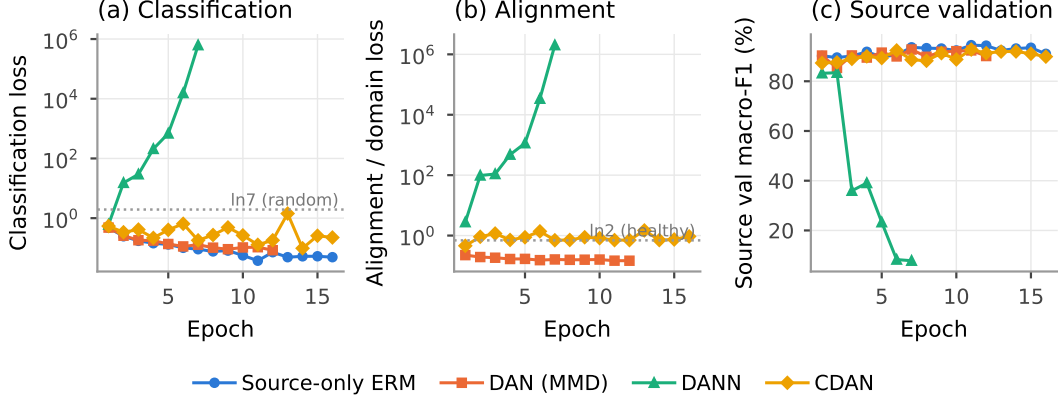


Figure 7: Training behaviour of the four methods, on logarithmic axes for the two losses. (a) Classification loss on the labelled source batches. The dotted line marks $\ln 7 = 1.946$, the value a seven-class classifier attains by predicting uniformly, so a curve above it is assigning less probability to the correct class than random guessing would. (b) Alignment loss: the MMD penalty for DAN and the domain-classification loss for DANN and CDAN. The dotted line marks $\ln 2 = 0.693$, the value a domain discriminator attains when it genuinely cannot distinguish the two domains, which is the equilibrium adversarial alignment aims for. CDAN remains close to this value throughout, whereas DANN departs from it from the second epoch onward. (c) Mean source-validation macro-F1, the quantity used for checkpoint selection. DANN’s losses diverge during epoch 2 while its source validation is still at its maximum, so the collapse is visible internally one epoch before the selection signal reflects it.

source-domain accuracy, as the source macro-F1 gained 7.46 points; however, the reduced alignment strength impacted the target macro-F1, decreasing it by 4.11 points. The pattern repeats when we further reduce the alignment strength, which further increases the model accuracy on source classification; however, it progressively damages target recognition, decreasing target macro-F1 by another 12.19 points. Interestingly, while we have these variations in target F1 score as we decrease the alignment strength, source and target separability remains around 100%. Therefore, in this scenario, the stronger or weaker adversarial objective failed to produce any measurable domain-invariant features.

In fact, all three DANN configurations remained well below the ERM target macro-F1 of 55.90%. So, the trade-off between low and high alignment strength had a substantial positive effect on target macro-F1 within the DANN variants; however, stronger alignment destabilized the model’s training, and a smaller value further deteriorated the target macro-F1.

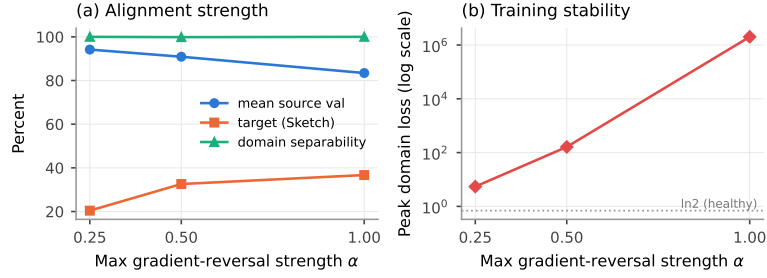


Figure 8: Controlled study over the maximum gradient-reversal strength $\alpha_{\max} \in \{0.25, 0.5, 1\}$ in DANN, with every other setting held fixed. Source macro-F1, target macro-F1 and domain separability are shown on a common percentage axis, with the Source-only ERM target macro-F1 marked for reference. Increasing alignment strength raises target macro-F1 and lowers source macro-F1, while separability stays at approximately 100% at all three settings, so the mechanism the objective is intended to produce is not observed at any strength. All three settings remain below the Source-only baseline on the target.

3 Task 3: Domain Generalization

3.1 Experimental Setup

Table 4: Source-side results for ERM, DAN-DG and SAM, with the two additional λ_{DG} settings of the controlled study below the rule. Each cell gives accuracy / macro-F1 (%) on the held-out validation split of that source domain. Worst is the lowest per-domain value of each metric; the weakest domain is Art Painting for every model except DAN-DG ($\lambda_{DG} = 10$), where it is Cartoon. These are the only quantities available for model selection, and checkpoints were selected on mean macro-F1.

Method	Photo	Art	Cartoon	Mean	Worst
ERM	96.11 / 95.53	92.68 / 92.64	94.67 / 95.37	94.49 / 94.51	92.68 / 92.64
DAN-DG ($\lambda_{DG} = 1$)	94.31 / 93.24	89.02 / 89.35	92.11 / 92.09	91.82 / 91.56	89.02 / 89.35
SAM ($\rho = 0.05$)	97.31 / 96.77	92.44 / 92.34	95.74 / 96.23	95.16 / 95.12	92.44 / 92.34
<i>Controlled study: alignment strength λ_{DG} in DAN-DG</i>					
DAN-DG ($\lambda_{DG} = 0.1$)	95.51 / 94.67	91.46 / 91.24	91.68 / 92.17	92.89 / 92.69	91.46 / 91.24
DAN-DG ($\lambda_{DG} = 10$)	41.32 / 21.54	25.12 / 15.39	18.12 / 11.24	28.19 / 16.06	18.12 / 11.24

Table 5: Sketch results and the two source-side diagnostics. Sketch accuracy and macro-F1 are measured once on the complete Sketch domain after every Task 3 configuration was frozen; Δ columns are changes relative to ERM in percentage points. Sep. is the held-out accuracy of a multinomial logistic-regression probe ($C = 1$, 70/30 split, seed 6304) predicting Photo, Art Painting or Cartoon from frozen source-validation features; chance is 33.3%. The sharpness proxy is measured on a fixed batch of 32 validation images per source domain (96 in total): $\mathcal{L}$ is the validation cross-entropy, $\mathcal{L}_{+\epsilon}$ the cross-entropy after one normalised gradient-ascent step of radius 0.05, and Δ_{sharp} their difference.

Method	Sketch				Sep.	Sharpness proxy		
	Acc	Macro-F1	ΔAcc	ΔF1		$\mathcal{L}$	$\mathcal{L}_{+\epsilon}$	Δ_{sharp}
ERM	55.71	55.90	—	—	86.38	0.192	0.792	0.600
DAN-DG ($\lambda_{DG} = 1$)	68.44	72.37	+12.73	+16.46	80.07	0.217	0.849	0.631
SAM ($\rho = 0.05$)	64.55	67.64	+8.83	+11.74	86.38	0.073	0.214	0.141
<i>Controlled study: alignment strength λ_{DG} in DAN-DG</i>								
DAN-DG ($\lambda_{DG} = 0.1$)	71.70	72.63	+15.98	+16.73	81.40	0.150	0.887	0.737
DAN-DG ($\lambda_{DG} = 10$)	10.56	7.84	−45.15	−48.06	48.84	1.855	4.103	2.248

3.2 Source Validation as a Predictor of Sketch Performance

Good performance on the source domain does not guarantee good performance on the target domain, and this is visible via the results as well. A high mean source accuracy / macro-F1 is observed in all 3 methods, where the ERM baseline mean source accuracy / macro-F1 is 94.49 / 94.51; the worst-performing source domain scored 92.64 points. Similarly, the mean source accuracy / macro-F1 for DAN-DG (with $\lambda = 1.0$) and SAM (with $\rho = 0.05$) is 91.82 / 91.56 and 95.16 / 95.12, respectively. Their worst-performing source macro-F1 is 89.35 and 92.34, respectively. While establishing a possible link between the validation numbers of source domains and the test numbers of the target domains, ERM, as anticipated, performed worst in the target domain, whereas DAN-DG has the lowest mean source accuracy of 91.82 and best target accuracy/macro-F1 of 68.44/72.37. SAM has performed the highest mean source accuracy of 95.16, but still its target accuracy is 3.89 points lower than DAN-DG. This shows there is no linear relation between the source and target domain performances.

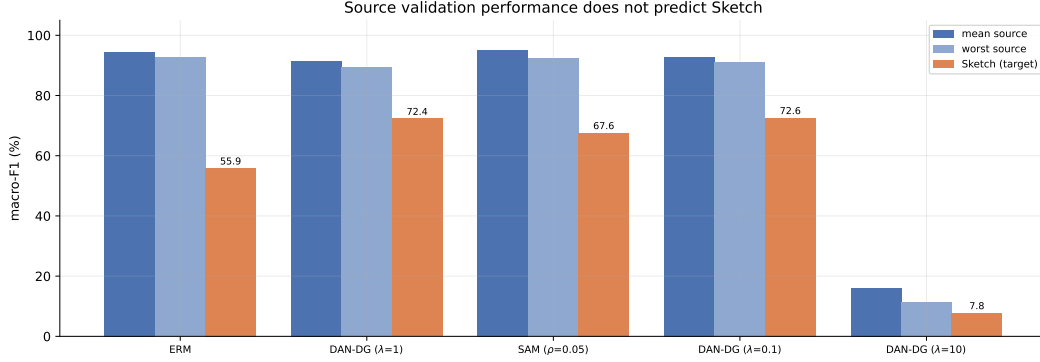


Figure 9: Mean-source, worst-source and Sketch macro-F1 for every Task 3 model, including the two additional λ_{DG} settings. The Sketch value is annotated above each orange bar. Source bars are close together for the four stable models, while their Sketch values span 55.9 to 72.6.

Table 6: Per-class Sketch accuracy (%) with the change against ERM in percentage points in brackets. Support is the number of Sketch images in each class. The final column is the target-aware DAN model from Task 2, measured against the same ERM checkpoint.

Class	Support	Task 3				Task 2
		ERM	DAN-DG ($\lambda=1$)	SAM	DAN-DG ($\lambda=10$)	DAN
dog	772	36.79	36.66 (−0.13)	30.18 (−6.61)	11.79 (−25.00)	63.60 (+26.81)
elephant	740	88.78	96.22 (+7.43)	98.51 (+9.73)	30.41 (−58.38)	87.16 (−1.62)
giraffe	753	23.51	44.89 (+21.38)	56.31 (+32.80)	0.00 (−23.51)	51.39 (+27.89)
guitar	608	83.06	95.56 (+12.50)	85.36 (+2.30)	0.66 (−82.40)	84.87 (+1.81)
horse	816	45.59	73.04 (+27.45)	55.64 (+10.05)	0.00 (−45.59)	47.67 (+2.08)
house	80	45.00	93.75 (+48.75)	81.25 (+36.25)	63.75 (+18.75)	100.00 (+55.00)
person	160	98.75	65.00 (−33.75)	70.00 (−28.75)	27.50 (−71.25)	97.50 (−1.25)

Therefore, it is difficult to establish that source validation performance reliably predicts performance on Sketch.

The details hidden by the aggregated source metrics come to light when looking at the class-level failures. ERM performs very poorly on Sketch for giraffe, dog, horse, and house, with accuracies of 23.51%, 36.79%, 45.59%, and 45.00%, whereas in the case of DAN-DG, 5 out of 7 classes gained points (house by 48.75 points, horse by 27.45, giraffe by 21.38, and guitar by 12.50). However, for the person, the performance declined by 33.75 points. Therefore, high aggregate source performance hides substantial target-specific class failures, particularly the degradation of person and the continued difficulty of dog recognition. These results are recorded in `per_class_sketch.csv`.

The art domain is recorded as the worst across all three methods.

3.3 Source-Domain Invariance and Class-Discriminative Information

DAN-DG made the observed source domains slightly less distinguishable, as shown by the source-domain separability score, which decreases to 80.07% compared with 86.38% under ERM. Its performance on Sketch also got better, and it reported an increase in macro-F1 from 55.90 to 72.37. These numbers appear to support the claim; however, the results change drastically when λ_{DG} is increased to 10. The separability score fell to 48.84%, but the macro-F1 on both the Sketch and the source just collapsed to 7.84% and 16.06%, respectively. Class-wise, 6 out of 7 classes lost substantial points: guitar fell by 82.40 points,

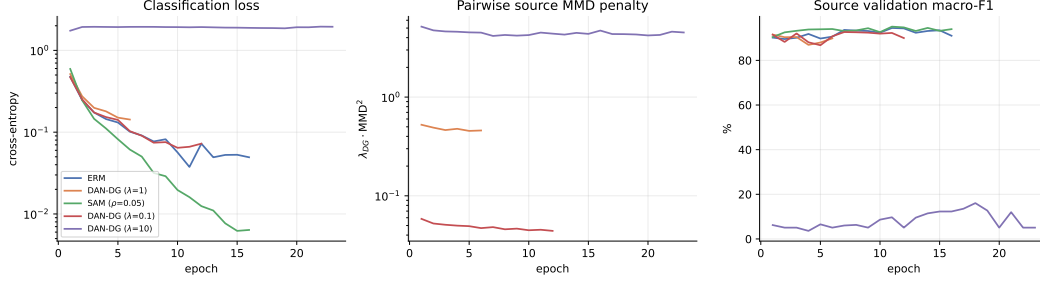


Figure 10: Training curves for all Task 3 models; ERM is the reused Task 2 run. (a) Classification loss on the source batches, on a logarithmic axis. (b) The weighted pairwise source MMD penalty $\lambda_{DG} \cdot \text{MMD}^2$ for the three DAN-DG settings. (c) Mean source-validation macro-F1, the checkpoint-selection signal. At $\lambda_{DG} = 10$ the classification loss stays between 1.74 and 1.95 for all 23 epochs, close to the $\ln 7 = 1.946$ of a uniform seven-class prediction, and source-validation macro-F1 never exceeds 16.06%. Curves end at the epoch where early stopping triggered; the selected checkpoints are epoch 11 (ERM), 1 (DAN-DG, $\lambda_{DG} = 1$), 11 (SAM), 7 ($\lambda_{DG} = 0.1$) and 18 ($\lambda_{DG} = 10$).

person by 71.25, elephant by 58.38, and both giraffe and horse reached 0% accuracy. Moreover, in the case of SAM and ERM, the separability is identical at 86.38%, yet SAM’s Sketch macro-F1 is 67.64 against ERM’s 55.90. Two models with exactly the same domain separability, yet a substantial performance gap. This suggests that excessive source alignment does remove domain-discriminative features, making domains more difficult to distinguish; however, at the same time, it destroys the core semantic structure required for classification. Therefore, moderate invariance among the domains partially improves Sketch recognition, while a stronger push results in loss of critical class-discriminative information, resulting in poor performance on the target domain.

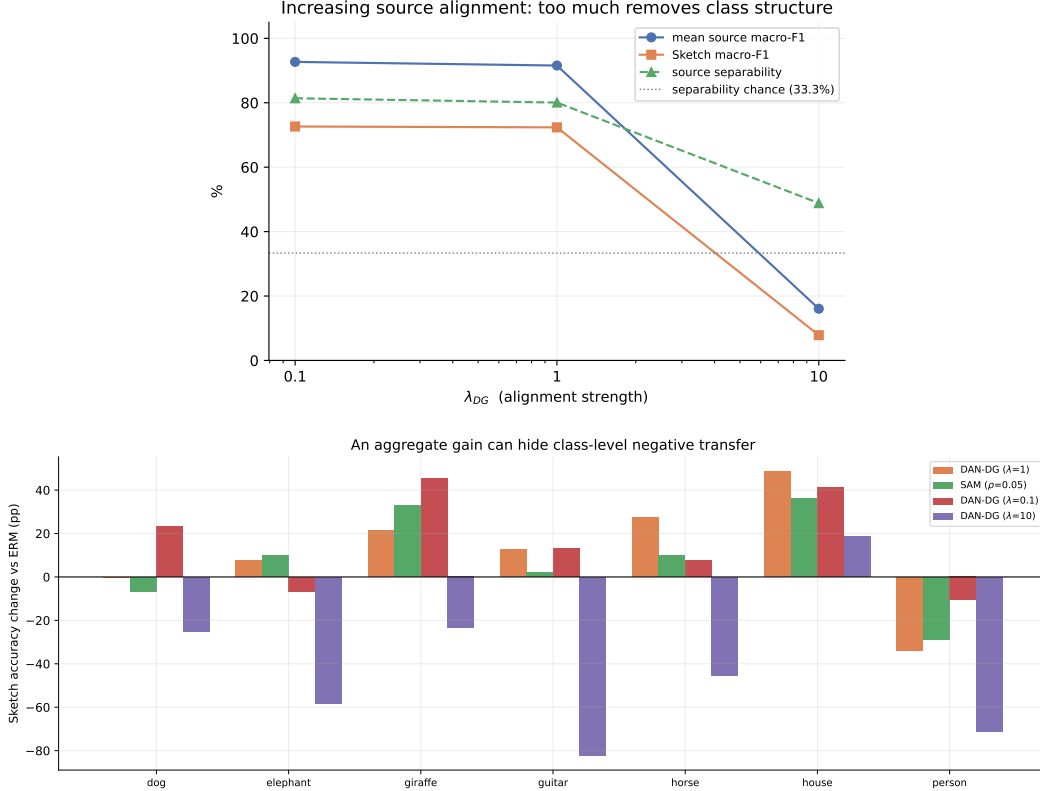


Figure 11: Top: controlled study over $\lambda_{DG} \in \{0.1, 1, 10\}$, with mean source macro-F1, Sketch macro-F1 and source-domain separability on a common percentage axis; the dotted line is separability chance (33.3%). Moving from 0.1 to 1 changes all three quantities by less than 1.5 points, while $\lambda_{DG} = 10$ pushes separability to 48.84% and collapses both source and Sketch macro-F1. **Bottom:** per-class Sketch accuracy change against ERM for each Task 3 model. At $\lambda_{DG} = 10$ only **house** improves.

3.4 Local Stability and Sketch Performance

A lower common sharpness proxy under given perturbation partially supports the claim about local stability, but it does not establish that the overall loss surface is flatter and will generalize better on any domain shifts. The model trained with SAM perturbation did appear to be a little less sensitive to any minor change in its weights. SAM’s loss increased by just 0.141, as compared to ERM and DAN-DG, which reported a change of 0.600 and 0.631, respectively. Therefore, the local stability ranking is

$$\text{SAM} < \text{ERM} < \text{DAN-DG}.$$

However, it does not establish that the local stability ranking enables better performance on the target domain, because on the target domain the performance of each model is

$$\text{DAN-DG} > \text{SAM} > \text{ERM}.$$

DAN-DG reported the highest sensitivity to a minor perturbation; however, its Sketch macro-F1 is the highest (72.37%), whereas SAM is much flatter as per the proxy ranking, but it reaches 67.64%. Nevertheless, SAM does improve over ERM by 8.83 accuracy points and 11.74 macro-F1 points, suggesting that local parameter stability may contribute to some extent to better generalization, but it is difficult to confirm that in every setting SAM will result in better performance on target domains.

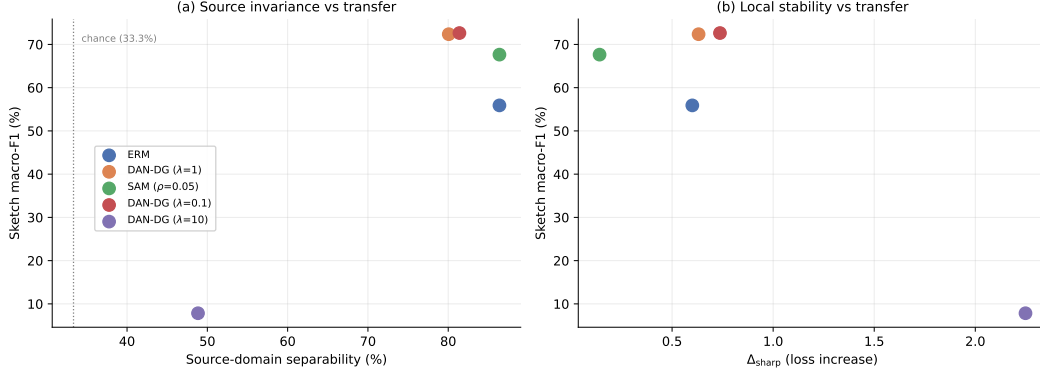


Figure 12: The two source-side diagnostics plotted against Sketch macro-F1. (a) Source-domain separability; the dotted line marks chance (33.3%). ERM and SAM share the same separability (86.38%) but differ by 11.74 points of Sketch macro-F1. (b) The sharpness proxy Δ_{sharp} ; lower means a smaller loss increase under the fixed perturbation of radius 0.05. SAM has the lowest value, but DAN-DG ($\lambda_{\text{DG}} = 1$) and DAN-DG ($\lambda_{\text{DG}} = 0.1$) reach higher Sketch macro-F1 with larger values.

3.5 Target-Aware DAN versus Target-Free DAN-DG

Both methods improve substantially over the shared ERM baseline:

Table 7: Target-aware DAN (Task 2) and target-free DAN-DG (Task 3) against the shared ERM checkpoint. Both use the same MMD implementation and RBF kernel construction.

Method	Training alignment	Sketch accuracy	Sketch F1
ERM	None	55.71%	55.90%
Task 2 DAN	Source versus unlabeled Sketch	67.80%	66.84%
Task 3 DAN-DG	Pairwise observed sources only	68.44%	72.37%

Both DAN and DAN-DG displayed comparatively better performance on the target domain as compared to ERM. DAN used unlabeled Sketch data to align the source features and improved over ERM by 12.09 accuracy points and 10.93 macro-F1 points. DAN-DG, which never had access to the target data (sketches) during training, improved by 12.73 accuracy points and 16.46 macro-F1 points. DAN-DG exceeds target-aware DAN by 0.64 accuracy points and 5.53 macro-F1 points.

This suggests that access to unlabeled Sketch data did not provide a substantial empirical advantage. Instead, it highlights that DAN-DG was able to learn the cross-domain invariances, which resulted in source-target alignment. However, this does not establish that unlabeled Sketch data has no value, and several limitations prevent this from being a complete causal attribution.

The whole experiment is limited to a single seed, 6304, a different seed might have produced different results.

DAN and DAN-DG work on different distributions; the DAN objective aligns source and target domain features given by $\mathcal{L}_{\text{align}}^{\text{DAN}} = \text{MMD}^2(F(X_P \cup X_A \cup X_C), F(X_S))$, whereas DAN-DG tries to learn invariances among the source domains:

$$\mathcal{L}_{\text{align}}^{\text{DAN-DG}} = \frac{\text{MMD}^2(P, A) + \text{MMD}^2(P, C) + \text{MMD}^2(A, C)}{3}.$$

In DAN, the pooled comparison between the source and target can hide the differences between the individual source domains; however, the pairwise terms in DAN-DG ensure that these differences are accounted for.

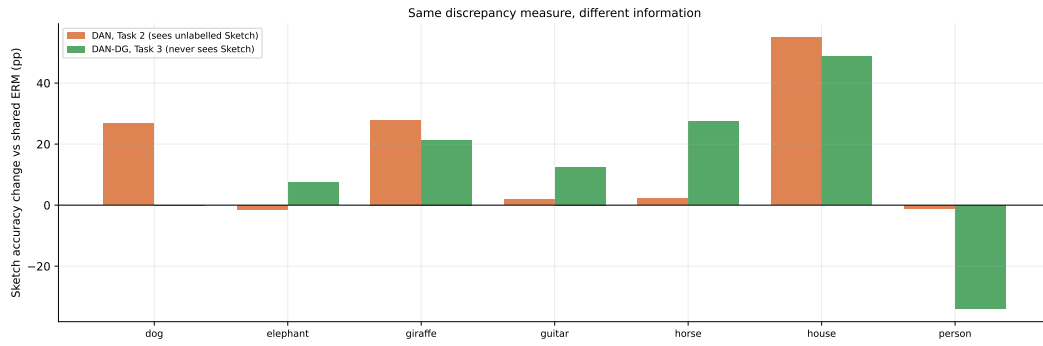


Figure 13: Per-class Sketch accuracy change against the shared ERM checkpoint for target-aware DAN (Task 2) and target-free DAN-DG (Task 3). Both improve **giraffe**, **guitar**, **horse** and **house**. They diverge on **dog** (+26.81 for DAN, -0.13 for DAN-DG) and **elephant** (-1.62 against +7.43), and both lose accuracy on **person**, DAN-DG by far more (-33.75 against -1.25).